HIGH-DISK WATERMARK

The high-disk watermark threshold is set to 90% by default, which protects the node's memory from falling lower than 90% of available disk memory. When this watermark hits, Elasticsearch makes every effort to move (relocate) shards from this failing node to other nodes where the memory is adequate.

Similar to the low-disk watermark, this is a dynamic setting, so we can use the cluster settings API to amend the threshold. The next listing decreases the high-disk-space watermark to 85% from the default 90% to make sure an alert is published when the line is crossed. Note that we use an alternate way of expressing 85%: 0.85.

Listing 15.15 Setting the high-disk watermark on the cluster

```
PUT _cluster/settings
{
  "transient": {
    "cluster.routing.allocation.disk.watermark.high":"0.85"
  }
}
```

FLOOD-STAGE-DISK WATERMARK

When the high-disk watermark is crossed, Elasticsearch waits for one more threshold, the flood-stage-disk watermark, before it enters panic mode. The default flood-stage watermark is 95%: if disk memory on the node falls below 95%, Elasticsearch raises this alert and makes all the indexes on this node non-writable. Any shards allocated on the node with the flood-stage watermark enabled turn into read-only shards.

As soon as disk space is available and disk usage doesn't tip the watermark threshold, the flood-stage watermark is reset. As with the other watermarks, we can (re)set the flood-stage watermark's default threshold to suit our requirements. In this listing, we turn down the knob on the flood-stage watermark.

Listing 15.16 Setting the flood-stage-disk watermark on the cluster

```
PUT _cluster/settings
{
  "transient": {
    "cluster.routing.allocation.disk.watermark.flood_stage":"90%"
  }
}
```

We can invoke `GET _cluster/settings` to fetch all the watermark settings to make sure they are set on the cluster:

```
"transient": {
    "cluster": {
      "routing": {
        "allocation": {
          "disk": {
            "watermark": {
              "low": "80%",
```